# Problem Challenge 1: Permutation in a String (hard)

## Problem Statement

Given a string and a pattern, find out if the **string contains any permutation of the pattern.**

**Permutation** is defined as the re-arranging of the characters of the string. For example, “abc” has the following six permutations:

1. abc
2. acb
3. bac
4. bca
5. cab
6. cba

If a string has ‘n’ distinct characters, it will have *n*! permutations.

**Example 1**:

```
Input: String="oidbcaf", Pattern="ABC"
Output: true
Explanation: The string contains "bca" which is a permutation of the given pattern.
```

**Example 2**:

```
Input: String="odicf", Pattern="dc"
Output: false
Explanation: No permutation of the pattern is present in the given string as a subs
tring.
```

**Example 3**:

```
Input: String="bcdxabcdy", Pattern="bcdyabcdx"
Output: true
Explanation: Both the string and the pattern are a permutation of each other.
```

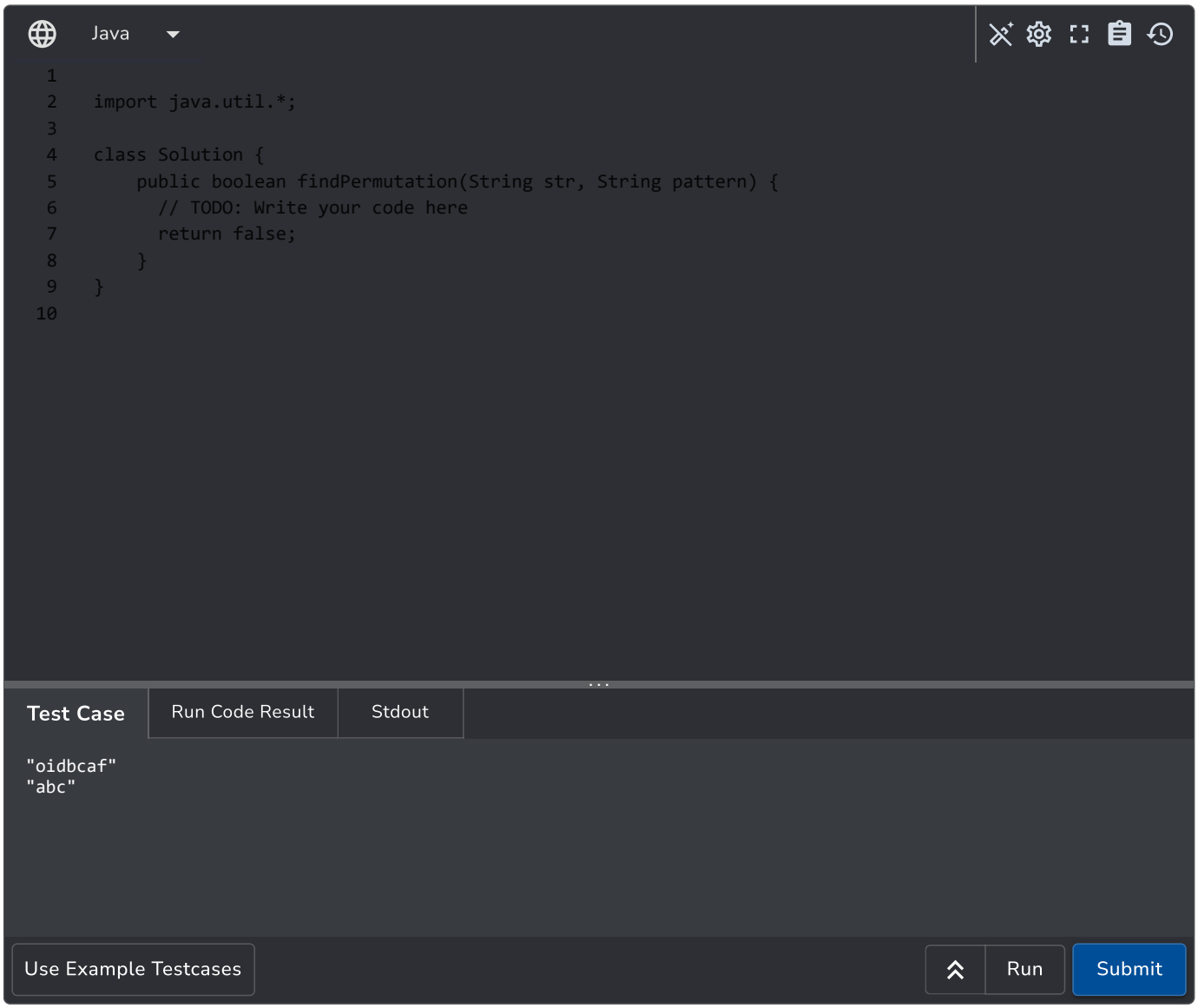
**Example 4**:

```
Input: String="aaacb", Pattern="abc"
Output: true
```

```
Explanation: The string contains "acb" which is a permutation of the given pattern.
```

## Try it yourself

Try solving this question here:





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Solution: Problem Challenge 1: Permutation In A String

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